

Impacts of quantum computing on artificial intelligence

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ABSTRACT

Quantum computing emerges as a new technological paradigm capable of significantly transforming Artificial Intelligence (AI) by providing computational power far beyond classical computing. This paper presents a systematic literature review aimed at investigating the impacts of quantum computing on AI, with a focus on information security and business environments. Recent studies were analyzed, highlighting benefits such as faster processing of large-scale data, advancements in fields such as healthcare, energy, industry, and sustainable development, as well as improvements in machine learning techniques. On the other hand, relevant challenges are identified including security vulnerabilities, the potential to break cryptographic algorithms, current technological limitations, quantum errors, scalability issues and difficulties in quantum software engineering. The study also discusses emerging solutions, such as post-quantum security models and blockchain applications. It is concluded that, despite its transformative potential, the adoption of quantum computing in AI depends on overcoming several technical, structural, and security challenges, making it a promising field for future research.

Keywords:

Quantum computing; Artificial intelligence; Information security; Cryptography; Business.

SUMMARY

Quantum computing emerges as a new technological paradigm capable of significantly transforming artificial intelligence (AI) by offering far superior computational power compared to classical computing. This article presents a systematic literature review aimed at investigating the impacts of quantum computing on AI, with an emphasis on aspects related to information security and the business environment. Recent studies highlighting benefits such as accelerated processing of large volumes of data, advancements in areas like health, energy, industry, and sustainable development, as well as improvements in machine learning techniques, were analyzed. On the other hand, significant challenges are highlighted including security vulnerabilities, the possibility of cryptographic algorithms breaking, current technological limitations quantum errors, scalability problems, and difficulties in quantum software engineering. The study also addresses emerging solutions, such as post-quantum security models and blockchain applications. It concludes that, although quantum computing represents great transformative potential for AI, its adoption still depends on overcoming several technical, structural, and security challenges, making it a promising field for future research.

Keywords

Quantum computing; Artificial intelligence; Information security; Cryptography; Business.

1. INTRODUCTION

Quantum computing will enable the development of computers far more powerful than those traditional computing, allowing processes that would take years to be completed in just a few hours or minutes. Combined with artificial intelligence (AI), numerous benefits for society could arise, such as improvements in industrial processes, rapid and efficient interpretation of large amounts of data, revolutionizing science, industry and business. On the other hand, extremely robust AI could weaken current encryption algorithms, increasing security and privacy problems that are already significant in traditional computing. One example is the Shor algorithm, which can break encryption systems (Hayward, 2008).

Quantum computers utilize properties of quantum physics to enable the use of qubits (also known as quantum bits) instead of traditional bits. Therefore, quantum computing is much more efficient than classical computing, currently used, allowing for complex applications with superior performance (Moran, CC, 2019). Algorithms that would take months or even years to run could be executed in just a few hours on a quantum computer.

Classical (current) computing uses the bit as the smallest unit of information. A bit can be represented by the numbers 0 or 1, which represent possible states. In quantum computing, qubits—linear combinations of both states—are used instead of bits, enabling much more logical states. This is why quantum computing has greater computational power.

The computing power of quantum computers was demonstrated in 2012 by Preskill, J. (2012): a theoretical physicist who discussed how a quantum computer can outperform a classical computer in solving a specific problem. This gave rise to the term "Quantum Supremacy". Despite the potential benefits, the high computing power of quantum computing could make it possible to break encryption algorithms, which would cause new problems or exacerbate existing ones related to privacy and security worrying companies, governments, and organizations. Because it has much greater computing power, it can break security algorithms, e.g., a 2048-bit RSA key can be broken in 8 hours if 20 million physical qubits are available (Ravi et al 2022).

The development of more efficient artificial intelligence (AI) applications may be made possible by the creation of quantum computers; the computational power of classical computing tends to be insufficient for future, more complex applications, starting with...

According to Moore's Law¹, which, according to many experts, is at its limit (Pollie, 2021) AI can be impacted by quantum computing by enabling the resolution of complex problems very quickly, something that is not possible with classical computing. The unknowns regarding the impacts of quantum computing on AI are numerous and could bring both harm and benefits. Therefore, there is an urgent need to identify research on the possible positive and negative impacts that quantum computing may bring and how artificial intelligence can enhance them.

The objective of this work is to investigate the possible impacts of quantum computing on artificial intelligence, related to security and information processing via the internet in the business environment of organizations. To this end, some specific objectives are pursued:

- Conduct a literature review on the concepts involved in quantum computing;
- Conduct a systematic literature review to identify the potential positive and negative impacts of quantum computing on AI and security;
- To identify through a literature review, possible solutions under development for preventing the impacts of AI on quantum computing and security.

This article is organized starting with this introduction. Section 2 presents the methodology used to develop this work; Section 3 discusses the results of the literature review; Section 4 presents the conclusion, and Section 5 presents the references.

2. METHODOLOGY

The methodology used in this Systematic Literature Review was based on Kitchenham (2004). To achieve the objectives described in the Introduction, section 1, a search for articles was conducted in databases and those selected were analyzed following these steps:

- Identify keywords;
- Combine terms to obtain the candidate string;
- Search with candidate strings using the following repositories: ACM Digital Library² and IEEE Xplore³;
- Evaluate the results obtained by the candidate string.

Therefore, the search string was defined based on the objective defined in Section 1 as being:

"(quantum computing) AND (artificial intelligence OR AI) AND (security OR information security) AND (business)"

The following inclusion criteria were defined for the selection of articles:

- articles published from 2019 onwards;
- articles that contain in the title or abstract at least one of the words used in the string;

- articles that address the objectives of this work;
- articles related to difficulties, characteristics, benefits and challenges for the application of quantum computing in AI;
- Availability of the text in English or Portuguese at no cost;
- publication in conference proceedings or scientific journals.

The exclusion criteria used were:

- articles whose full text could not be accessed;
- Proposed solutions that were not designed for the context of quantum computing in information security and intelligence. Artificial;
- work focusing on areas not directly related to the research questions;

- For duplicate submissions, those with the most recent publication will be selected.

The basis for the studies was obtained using the search string in the previously defined repositories. A total of 51 articles were returned using the search string. The articles found in the digital libraries were counted, with 44 in the ACM Digital Library and 7 in IEEE Xplore. After applying the selection criteria, articles that could not be accessed were rejected, applying the first exclusion criterion, and the remaining articles underwent a reading of the abstracts to apply the other criteria. It was observed that not all articles are related to the objective of this research. In the end, after applying the inclusion and exclusion criteria, 46 articles were rejected and 5 were accepted.

3. RESULTS

Software engineering for quantum computing presents several challenges e.g., in quantum software testing, model-driven engineering (MDE) quantum computing, quantum programming paradigms, quantum software architectures, quantum software development processes, and quantum artificial intelligence (Murillo et al., 2025). Some authors also discuss the potential benefits that quantum computing can provide. These benefits and challenges will be discussed below.

3.1 Possible benefits provided by Quantum Computing

By using qubits, quantum computing is much faster than traditional computing (Moran et al. 2019). This could enable the execution of a much more powerful AI capable of performing operations that would be unfeasible on current machines. According to Nedungadi et al. (2024), the use of Artificial Intelligence and big data can contribute to advances in three UN Sustainable Development Goals: SDG 3 good health and well-being, SDG 7 affordable and clean energy, and SDG 9 industry, innovation and infrastructure.

- Many AI and big data tools can be used to improve diagnostics. By analyzing large volumes of

¹ Note that the number of transistors in processors doubles every two years, increasing computing power. Pollie, R. (2021).

² <https://dl.acm.org/>

³ <https://ieeexplore.ieee.org/Xplore/home.jsp>

Information, such as medical records, images, and genetic information, allows AI systems to find patterns, enabling rapid diagnoses and decision-making. The application of AI in mental health provides techniques such as virtual assistants and chatbots to offer evidence-based recommendations for personalized treatments. Machine learning tools for analyzing medical records are suggested to demonstrate AI's ability to increase diagnostic accuracy and treatment effectiveness in orthopedics, contributing to goal 3 (health and well-being). Deep learning models can automatically learn and extract features to analyze medical data such as X-rays, MRI scans, and CT scans to detect anomalies with high accuracy.

- A land big data techniques can optimize oil and gas extraction, a key component of the energy sector. The use of neural networks and evolutionary algorithms improves the extraction process, ensuring more efficient energy production. Energy-efficient hardware designs for AI and machine learning, focused on improving the power and energy efficiency of information processing systems, contribute to goal 7 (clean and affordable energy);

- Machine learning can monitor machine health by analyzing images and videos. This insight, based on rapid problem detection and diagnosis, minimizes downtime and improves overall equipment maintenance efficiency, aligning with goal 9 (industry, innovation, and infrastructure). Cyber-physical industrial systems can be improved through the use of machine learning and multi-agent systems. Automation and efficiency are crucial for goal 9.

Quantum computing can be used in the creation and editing of films with quantum image processing (QIP) and quantum audio processing (QAP), as presented by Ilyas et al. (2019). Quantum image processing (QIP) is an emerging technology focused on extending tasks from conventional image processing. It is developed to "use quantum computing technologies to capture, manipulate, and retrieve quantum images in different formats and for different purposes".

Due to some of the inherent properties of quantum computing and parallelism, QIP technologies are expected to offer capabilities that are still unmatched by their traditional equivalents. QIP seeks to search for patterns in images. In the literature, there are three groups of quantum image processing:

1. Optics-based quantum imaging (OQI): investigates the ultimate limits of optical imaging that are permitted by the laws of quantum mechanics;
2. Quantum-assisted digital image processing (QDIP): focuses on exploring properties responsible for the power of algorithms to improve certain tasks;
3. Classically inspired quantum image processing (QIP): applications that derive their inspiration from potential quantum hardware and focus on research that concentrates on extending

Deep learning is a subset of machine learning technology that uses multi-layered neural networks, called deep neural networks, to simulate the power of decision-making.

QDIP tasks and applications for quantum computing framework.

3.2 Challenges for Quantum Computing

Despite the benefits, many challenges still exist for quantum computing. According to (Saki et al., 2021), there are several vulnerabilities and attack models in quantum computing:

- Qubit lifetime and errors: Qubits in NISQ (Noisy Intermediate-Scale Quantum) devices suffer from short lifetimes, faulty gates and measurement operations. A short qubit lifetime implies a spontaneous loss of the qubit's state (saved data), which is called quantum decoherence, e.g., a qubit in state "1" loses energy when interacting with the environment and changes its state to "0". This phenomenon is called relaxation;
- Limited native gates: a quantum program or circuit can be built with any high-level quantum gate. However, NISQ devices only support a few native gates, e.g., IBM machines only have four single-qubit gates and one two-qubit gate;
- Coupling problem: NISQ devices, especially superconducting qubit-based computers, have limited connectivity between qubits known as the coupling problem. This limited connectivity prevents two qubit gates from connecting to each other;
- Cloud access: There are several quantum computing processors and simulators in the cloud. If cloud services are the only option, several security and privacy issues may arise. For example, a malicious entity on the cloud server could report incorrect error rates;
- Coding problems: the construction of a quantum circuit can reveal sensitive information about the problem. The article discussed a vulnerability using a MaxCut combinatorial problem that involves splitting a graph into two parts so that the maximum number of edges are cut;
- Use of unreliable compilers: One of the important aspects of compiling quantum circuits is optimizing the circuit to improve circuit depth and reduce gate count. Several compilers are involved in offering optimization and fast compilation, even in large quantum circuits. The following factors will motivate quantum circuit developers to evaluate the unreliability of these compilers: (a) quantum circuit success from an optimized circuit is essential to obtain meaningful results from NISQ computers. A poorly optimized circuit may produce random outputs even if its functionality is identical; (b) lack of reliable compilers with the latest advances in optimization; (c) availability of efficient but unreliable compilers, which are developed to optimize circuit depth and gate count, compared to reliable compilers. These compilers may be hosted on local machines by third parties in cloud services to be performed for: (i) cloning/forgery, i.e., when a quantum circuit can be stolen or reproduced; and (ii) engineering.

The decision-making process of the human brain. So, the form of deep learning exists in most artificial intelligence (AI) applications.

5. Longest route of the circuit.

reverse, i.e., when sensitive aspects of a quantum circuit can be extracted;

- Vulnerabilities in quantum machine learning applications: Quantum machine learning (QML) is an emerging field that aims to develop quantum algorithms to perform conventional generative/discriminative machine learning tasks (e.g., classification, regression, etc.). Large-scale classical computing data can be loaded into a few qubits;

- Security attacks: Error injection by interference. Error interference can be exploited to launch error injection attacks in MTC (multi-tenant computing) environments, when two or more quantum programs run simultaneously on different sets of physical qubits;

- Scheduling attacks: quantum circuits are sent to quantum hardware via a cloud provider that allocates the hardware to the circuit. Here, the user has no visibility into the hardware to be allocated. The authors (Phalak et al., 2021) propose a new attack model where the user is allocated inferior quantum hardware instead of the desired one;

- Attacks on Quantum Machine Learning (QML): similar to attack models in classical machine learning algorithms, attacks on QML can be categorized into three dimensions: (i) attack during training or inference, (ii) type of information available to the attacker (e.g., knowledge of the QML internal models and algorithms or access to inputs and outputs), and (iii) attacker objectives (e.g., misclassification of certain inputs or affecting the overall reliability of the model).

The work of Ramalho et al. (2025) discusses categorized bugs (errors) found in projects related to quantum computing:

- Configuration bugs: These are problems related to configurations, such as incorrect file paths, directories, or the need to update external libraries;

- Data types and structures bugs: bugs involving undefined or incompatible data types and data structures, such as improper use of data structures;

- Missing error: occurs when exceptions are not thrown properly, causing crashes or errors;

- Performance bugs: problems that affect the stability, speed or responsiveness of the software, such as memory issues and infinite loops;

- Permission bugs or obsolescence: problems related to outdated calls or APIs, or incorrect API permissions;

- Program anomaly bugs: introduced when existing code is extended poorly implemented, leading to crashes or the return of incorrect values;

- Bugs related to code testing: problems in the test code due to addition or updating of test cases or to incorrectly executed tests;

- Database-related bugs: problems that occur with the application communicating with a database;

- Documentation: problems related to outdated documentation or missing information;

- GUI-related bugs: problems with graphical elements, such as layout issues in text boxes and buttons, which affect the usability and interpretation of quantum circuits;

- Misuse: errors caused by incorrect use of functions;

- Network bugs: problems related to connectivity or server issues, including cloud access in quantum components;

- Monitoring: bugs due to improper logs, such as incorrect log levels, excessive logs, or missing statements in logs.

3.3 Challenges for Quantum Software Engineering

In software engineering several disciplines can be affected, e.g., testing and processes. The following discusses some of the problems for software engineering found in this literature review.

I. Challenges for testing quantum software.

Murillo et al. (2025) discussed the following challenges for quantum software testing:

- Efficiency test: high resource consumption. These tests typically need to be run for each input/output, increasing the computational cost;

- Scalability testing: challenges related to efficiency determined by the number of qubits and quantum states;

- Simulators for real quantum computers: reliability is still lacking;

- Quantum Artificial Intelligence and quantum software testing: potential use of classical AI to test quantum. This can also help in testing classical software;

According to Ramalho et al. (2025), there are several challenges, such as testing and debugging quantum programs. This work discusses various testing approaches for quantum programs.

Combinatorial testing: has scalability issues such as the increase in qubits;

- Research-based testing: QuSBT, a test generation tool for quantum programs, uses genetic algorithms to create a test suite with a maximum number of failed tests;

- Fuzz testing: involves inputting data into programs. Its use in generating rare inputs for quantum programs to cause sensitive branches (code branching) and induce locks to discover defects was investigated. The goal is to use a gray-box approach to identify code operations and their resulting branches;

- Property-based testing: a structured method for automating tests using program specifications (Honarvar apud Ramalho, 2025). It has been studied as an alternative to mitigate the non-deterministic nature of quantum programs;

- Mutation testing: has two important roles in quantum programming (i) mutation operators can be used to create faulty versions of quantum programs, mitigating the lack of quantum bug repositories and (ii) benchmark testing of quantum programs to assess the quality of test applications of quantum programs.

- Metamorphic testing: an approach to using metamorphic relations, i.e., functions written to be executed directly on quantum computer, to test quantum programs. The authors use metamorphic rules, written as quantum functions and based on quantum program properties, to avoid or delay qubit measurement;

- Unstable tests: some studies focus on unstable tests. These are tests present in non-deterministic behaviors that sometimes succeed and sometimes fail;

- Test coverage: a coverage criterion is a rule or collection of rules that impose test requirements on a test suite.

II. Challenges for quantum computing oriented towards model-driven development (MDE).

According to Murillo et al. (2025), quantum computing has some challenges related to the development of model-based software:

- Interoperability: challenges related to the lack of standardized frameworks and development environments. Despite the existence of the Open Quantum Assembly Language (OpenQASM) standard, there are still several assembly-level languages, specific to each type of hardware or platform, which, according to the authors, create barriers to the development of quantum software;

- Platform independence: a major problem is the differences in the capabilities of Quantum Processing Units (QPUs). Some development platforms have different execution times and other optimizations specific to their own platform. This creates challenges for quantum software developers;

- Demand and manageability: quantum hardware introduces more complexity in manageability due to its limitations such as the accuracy of quantum operations, qubit duration in superposition and qubit connectivity, and the ability of qubits to interact with each other within a quantum processor;

- Training: transitioning from classical to quantum or hybrid software development. This is a multifaceted challenge, requiring the learning of new technical knowledge and the ability to integrate quantum principles with existing classical architectures.

- Specific modeling for quantum computing: unlike classical software, quantum software requires precise representations of quantum gates, circuits, and states, which necessitates specialized MDE languages, techniques, and tools capable of modeling the components;

- Development of high-level methodologies: crucial for bridging the gap between classical and quantum computing paradigms. This involves creating abstract modeling frameworks that encapsulate the complexity between the hybrid interactions of quantum and classical computing;

- Maintenance and evolution of quantum software: as quantum software becomes more complex, maintaining, disseminating and evolving it will present significant challenges. Future research could explore MDE approaches to predict the impact of changes in quantum software;

- Intelligent code generation: By automating the generation of quantum code from high-level models, developers will be able to focus more on solving the problem than on the complexities of quantum programming languages.

III. Challenges of paradigms in quantum programming (Murillo et al. (2025):

- Circuit complexity: one of the main difficulties in quantum computing, especially in the implementation of algorithms that require sophisticated operations;

- Reuse of quantum software: essential for reducing the complexity of developing quantum algorithms and circuits. Developing complex quantum circuits may require extensive knowledge of quantum mechanics and quantum states;

- Abstractions for quantum software: essential for increasing the usability and flexibility of quantum programming languages. The potential of quantum computing lies in its ability to model and simulate complex quantum systems, such as those in chemistry and physics.

IV. Challenges in quantum software architectures, Murillo et al (2025):

- Quantum software architectural decisions: require a comprehensive understanding of both the implementation details and technical choices that shape the development of a robust quantum computing system;

- Developing standards for hybrid systems: integrating classical and quantum computing is essential to ensure scalability and flexibility. At the low level, the focus should be on establishing standards to promote modularity, reusability, and efficiency. At the high level, architectural patterns need to be developed to manage services and the execution of workflows that encompass both quantum and classical systems;

- Empirical evidence for the application of standards: necessary to bridge the gap between theory and practice in industrial and hybrid quantum computing;

- Evolution of hybrid software architectures: crucial for understanding the challenges and complexities involved in integrating quantum systems with classical systems.

V. Challenges for software development processes quantum, Murillo et al (2025):

- Hybrid interactive software development: interactive development models that have proven effective in managing complexity and increasing the flexibility of classic software projects should be adapted to hybrid systems.

Hybrid software development. This includes models that accommodate rapid changes in quantum technology and provide frameworks for developing both classical and quantum components;

- Risk management: essential for the development of larger more complex hybrid software systems;
- Project management: focuses on the operational aspects of quantum software, requiring the integration of DevOps or similar agile paradigm into the software development lifecycle.

3.4 Challenges Related to Artificial Intelligence and Blockchain

For quantum artificial intelligence, some challenges are listed by Murillo et al. (2025):

- Optimization of quantum circuits: Quantum circuits are very complex and require optimization to function efficiently on current quantum hardware. While AI has shown potential in optimizing classical software, its application to quantum circuits presents significant challenges. AI must take into account quantum characteristics such as the limitation on the number of qubits;
- Development of hybrid workflows for quantum AI: the current state of quantum computing requires hybrid workflows involving classical and quantum components. The challenge lies in integrating AI techniques with hybrid classical and quantum computing systems;
- Error mitigation and correction: Quantum systems are problematic, and error correction is one of the main challenges. While AI automates some aspects of error detection and correction in classical computing, the nature of quantum errors, such as gate errors, requires new approaches.
- Scalability of AI-assisted quantum software development: As quantum systems become scalable, the software and algorithms that control them must also be. AI has the potential to accelerate quantum software development but ensuring that the AI-driven development process is scalable and efficient is a major challenge.

Solutions for Post-Quantum Blockchain Consensus (PQBC) are presented by Gomes et al. (2023):

- Consensus based on random quantum numbers: Random numbers are important in Blockchain applications. Verifying that the generated number is random is important for the security of the applications;
- Quantum entanglement and consensus based on quantum measurement: occurs when a quantum system generates a set of small particles that share quantum states. However, this new entangled state only exists together, and it is impossible to recreate these particles independently. That is, a quantum entanglement state exists if and only if it is impossible to represent it as a vector of qubits;
- Consensus based on quantum key distribution: quantum key distribution creates a secure channel where peers

(i.e., network participants) can share secret keys. Only peers on the network know these keys;

- Consensus based on quantum distributed processing: Quantum distributed processing assumes that quantum computers will be faster than classical ones. Thus, systems based on quantum computing will be secure against quantum attacks. However, the principles of quantum entanglement, quantum measurement, quantum key distribution, and quantum random number generation are used in distributed processing. The main goal of quantum distributed processing is to be secure and reliable against quantum computers;
- Consensus based on diagonal networks: solutions designed to be safe and built on vectors and diagonal networks;
- Consensus based on multivariate polynomial equations: this type of equation is considered secure against attacks from classical and quantum computers;

According to Gomes et al. (2023), there are four stages for Post-Quantum Blockchain Consensus solutions to achieve consensus:

1. Leader election: Blockchain participants will decide who will lead the generation of the block;
2. Block generation: miners create a valid block consisting of a packet of transactions, generating new quantum code (hash) and communicating with a new block for peer verification;
3. Block validation: the leader communicates with a block to verify pairs;
4. Chain update: a new block is inserted into the Blockchain and it is updated to the new version.

Based on the analyzed work, the consequences of quantum computing on artificial intelligence are surrounded by challenges, both positive and negative aspects. Several challenges exist within quantum software engineering itself, such as testing (efficiency and scalability tests), interoperability (lack of frameworks and development environments), maintenance, and evolution of quantum software. Further challenges arise for quantum artificial intelligence – e.g., optimization of quantum circuits, workflow development, error mitigation and correction – impacting its development.

Many problems and challenges in quantum computing analyzed in this article resemble those of classical computing. Since quantum computing is much more powerful than classical computing, bugs could theoretically be amplified. On the other hand, AI in quantum computing – due to its computational power – could bring positive aspects such as: benefits in public health, through the analysis of large volumes of medical information and decision-making; optimization of oil and gas extraction; and detection and diagnosis of problems in industry, contributing, respectively, to the UN's Sustainable Development Goals 3 (health and well-being), 7 (clean and affordable energy), and 9 (industry, innovation and infrastructure).

7 DevOps is a software development practice with tools that unite software development teams.

(Dev) and IT operations (Ops) to deliver applications and services faster.

referenced (eg, "[Robertson, personal communication]").

4. CONCLUSION

The main areas where quantum computing is expected to have the greatest positive impact are:

- Renewable Energy: Development of new materials (such as lightweight batteries for cars and airplanes) and new catalysts (Ukpabi, 2023);
- Pharmacy: Vaccine and drug development often much faster than current development (Ukpabi, 2023);
- Medicine: Diagnoses, research, treatments and interventions (Flöther, 2023);
- Industry: Various improvements, increasing production efficiency. Conducting vehicle accident simulations and training algorithms used in autonomous driving software (Ukpabi, 2023);
- Film industry: Creating and editing films with capabilities superior to their traditional counterparts (Iliyasu, 2019);

Challenges are expected in relation to:

- Interoperability: lack of standardized frameworks (Murillo, 2025);
- Circuit complexity: one of the main difficulties in quantum computing, especially in the implementation of algorithms that require sophisticated operations (Murillo, 2025);
- Vulnerabilities in QML applications: large-scale classical computing data can be loaded into a few qubits, potentially exacerbating problems (Saki, 2021).

It is concluded that the creation of a quantum computer will bring numerous benefits to industry, leisure, cinema, entertainment, medicine, and public health. However, since quantum hardware does not yet exist, quantum computing is still a very theoretical concept with several challenges related to modeling, evolution, software maintenance and vulnerabilities that can be aggravated due to high computational power; challenges in testing and debugging, bugs, lack of standardized frameworks and development environments; error mitigation and correction, circuit optimization, quantum entanglement, and the scalability of AI-assisted quantum software development.

Creating functional quantum hardware and solving its problems and challenges could enable the development of highly effective. It is capable of bringing numerous practical benefits to society.

This work presented a literature review; however, few studies exist in the literature on issues related to quantum computing and artificial intelligence, which is a limitation. Although this research for scientific publications was carried out in two digital libraries of scientific articles recognized in the academic community, it is necessary to extend this research.

This research is being extended to other libraries in order to reach the largest possible number of articles that discuss the topic. As it is a relatively recent field, quantum computers are not yet available, and few studies exist.

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